Generating Chest X-rays with a Convolutional Variational Autoencoder

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Abstract

We implement a convolutional variational autoencoder (VAE) [1] in PyTorch and train it on the pneumonia subset of the Kaggle chest X-ray dataset. We report hyperparameters, the loss curve, Inception Score (IS) and Fréchet Inception Distance (FID) via PyTorch-Ignite, and samples. The model reaches an FID of 129.43 against the test set (real held-out training images score 20.5, uniform noise 364.3) and an IS of 1.579.

1 Setup

Data. We use the chest X-ray pneumonia dataset [2]. The repository contains the PNEUMONIA class only: 3875 training and 390 test images. Each image is converted to a single grey channel, resized to 64×64 for compute reasons, and scaled to $[0, 1]$. No augmentation is used. All code (`VAE/`, `train.py`, `eval_model.py`, `data.py`, `main.py`, `train_gpu.sbatch`) is submitted with this report.

Table 1 lists every setting; training ran on one NVIDIA RTX PRO 6000 Blackwell (1g.24gb MIG slice) GPU. Evaluation results are in Table 2.

2 Results

Training loss. Figure 1 shows the per-image training loss (reconstruction plus KL) over 100 epochs. The loss falls from 2562.3 after the first epoch to 2403.1 at the end, with a minimum of 2402.2 at epoch 97. About 88% of the decrease happens in the first ten epochs; the loss is dominated by the reconstruction term.

Inception Score and FID. Following the PyTorch-Ignite recipe [3], we decode one sample $z \sim \mathcal{N}(0, I)$ per real test image and update Ignite's `FID` and `InceptionScore` batch by batch. Images are replicated to three channels and upsampled to 299×299 for the ImageNet Inception-v3. FID uses the standard 2048-d pool features; Ignite's default `FID()` instead uses the 1000-d softmax outputs, giving a meaningless 0.115 here, so we pass the pool-feature extractor explicitly. For context, Table 2 also scores 390 real training images (20.5, roughly the floor at this sample size) and uniform noise (364.3) against the test set. Our 129.43 lies about a third of the way from the real-data floor to noise, consistent with the blurry samples. IS is close to its lower bound of 1 because Inception assigns nearly the same class distribution to every chest X-ray, so FID is the more informative metric.

Generated images. Figure 2 shows sixteen random samples. The model captures the global anatomy of a frontal chest radiograph: the dark lung fields, the bright mediastinum and spine, and the shoulder and rib outline. As is typical of a VAE trained with a pixel-wise likelihood, the samples are blurry and lack fine detail such as individual ribs or vessel markings. Higher resolution, a perceptual reconstruction term, or a tempered KL weight are natural next steps.

Setting	Value
Input	64×64, 1 channel
Latent dimension	128
Optimiser	Adam, lr 10^{-3}, default β
Batch size / epochs	32 / 100
Loss	BCE (sum) + KL, weight 1
Activations	LeakyReLU(0.2) enc., ReLU dec.
Normalisation	BatchNorm
Data loading	8 workers, shuffled

Table 1: Hyperparameters.

Metric (vs. 390 test images)	Value
FID ↓, VAE samples	129.43
real training images	20.5
uniform noise	364.3
IS ↑, VAE samples	1.579

Table 2: FID and IS (PyTorch-Ignite). Indented rows are reference points.

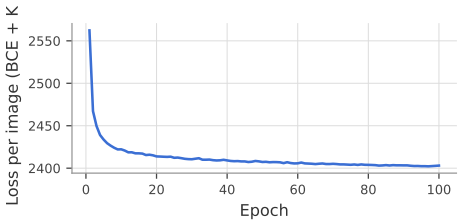


Figure 1: Training loss per image per epoch.

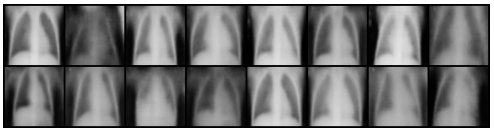


Figure 2: Sixteen samples decoded from $z \sim \mathcal{N}(0, I)$.

References

[1] D. P. Kingma and M. Welling. Auto-encoding variational Bayes. *ICLR*, 2014.

[2] D. S. Kermany et al. Identifying medical diagnoses and treatable diseases by image-based deep learning. *Cell*, 2018. `https://www.kaggle.com/paultimothymooney/chest-xray-pneumonia`.

[3] PyTorch-Ignite. GAN evaluation with FID and IS. `https://pytorch-ignite.ai/blog/gan-evaluation-with-fid-and-is/`.